

Customer Churn Prediction in the Telecom Industry

Identifying Churn Drivers Through Statistical Modelling and Logistic Regression

Business Problem Statement

Project Aim

Customer churn is a major challenge for telecom companies, as retaining existing customers is often more cost-effective than acquiring new ones. This project aims to identify customers who are likely to discontinue services, understand the factors contributing to churn, and support business decision-making through statistical analysis and predictive modelling.

About the Dataset

The dataset used in this project is the **IBM Telco Customer Churn Dataset**, which contains information about customers of a telecommunications company. The dataset is widely used for customer retention analysis and churn prediction studies.

Each row in the dataset represents an individual customer, while each column describes a specific customer attribute related to demographics, subscribed services, account information, and churn status.

Statistical Analysis and Predictive Modelling Using R

1. Data Import and Exploration

R code:

```
> dim(WA_Fn_UseC_Telco_Customer_Churn)
```

Output

```
[1] 7043 21
```

Result

The dataset consists of **7,043 customer records and 21 attributes**, providing comprehensive information for customer churn analysis and prediction.

R code:

```
> names(WA_Fn_UseC_Telco_Customer_Churn)
```

Output

```
[1] "customerID" "gender"      "SeniorCitizen" "Partner"
[5] "Dependents"  "tenure"      "PhoneService"  "MultipleLines"
```

[9] "InternetService" "OnlineSecurity" "OnlineBackup" "DeviceProtection"

[13] "TechSupport" "StreamingTV" "StreamingMovies" "Contract"

[17] "PaperlessBilling" "PaymentMethod" "MonthlyCharges" "TotalCharges"

[21] "Churn"

Result

The dataset consists of **21 variables** representing different aspects of customer information, including demographics, subscribed services, account details, billing information, and customer churn status.

The variables can be broadly categorized as:

- **Customer Demographics:** Gender, Senior Citizen, Partner, Dependents
- **Service Information:** Phone Service, Multiple Lines, Internet Service, Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, and Streaming Movies
- **Account Information:** Tenure, Contract Type, Paperless Billing, Payment Method, Monthly Charges, and Total Charges
- **Target Variable:** Churn

The presence of diverse customer, service, and billing-related variables provides a comprehensive foundation for identifying factors that influence customer churn and developing predictive models for customer retention.

2. Data Cleaning and Preprocessing

R Code:

```
> colSums(is.na(WA_Fn_UseC_Telco_Customer_Churn))
```

Output:

customerID	gender	SeniorCitizen	Partner	Dependents
0	0	0	0	0
tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity
0	0	0	0	0
OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies
0	0	0	0	0
Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges
0	0	0	0	11
Churn				
0				

R Code:

```
> churn_data <- na.omit(WA_Fn_UseC_Telco_Customer_Churn)
> dim(churn_data)
```

Output:

```
[1] 7032 21
```

Result

The missing value analysis revealed that **all variables are complete except for TotalCharges**, which contains **11 missing values**. Since the number of missing observations represents only a very small proportion of the dataset (11 out of 7,043 records), these records were removed during data cleaning to ensure the quality and reliability of the analysis.

The analysis indicates that the dataset is largely complete and suitable for statistical analysis and predictive modeling, with minimal impact from missing data.

3. Exploratory Data Analysis (EDA)

R Code:

```
> table(WA_Fn_UseC_Telco_Customer_Churn$Churn)
```

Output:

```
No    Yes
5174 1869
```

R code:

```
> prop.table(
+   table(WA_Fn_UseC_Telco_Customer_Churn$Churn)
+ )*100
```

Output:

```
    No    Yes
73.46301 26.53699
```

R code:

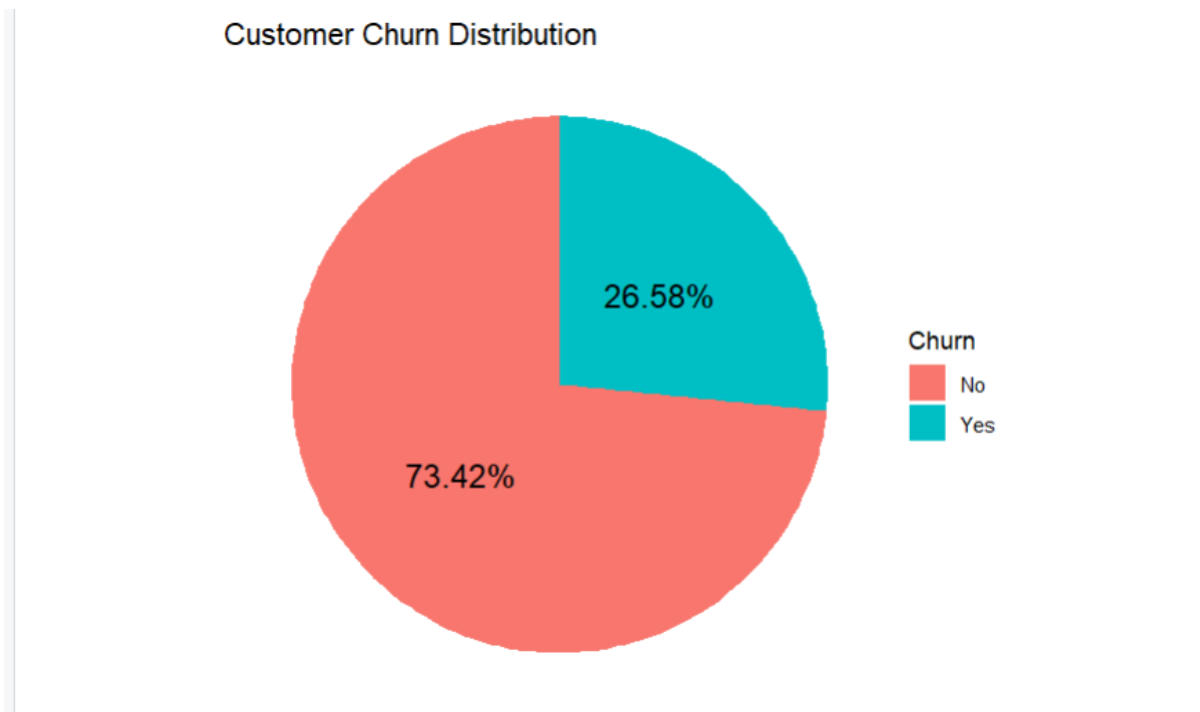
```
> library(ggplot2)
>
> # Create data frame
> churn_df <- as.data.frame(table(churn_data$Churn))
> colnames(churn_df) <- c("Churn", "Count")
>
```

```

> # Calculate percentage
> churn_df$Percentage <- round(churn_df$Count / sum(churn_df$Count) * 100, 2)
>
> # Pie chart with percentages
> ggplot(churn_df, aes(x = "", y = Count, fill = Churn)) +
+   geom_col(width = 1) +
+   coord_polar(theta = "y") +
+   geom_text(
+     aes(label = paste0(Percentage, "%")),
+     position = position_stack(vjust = 0.5),
+     size = 5
+   ) +
+   labs(title = "Customer Churn Distribution") +
+   theme_void()

```

Output:

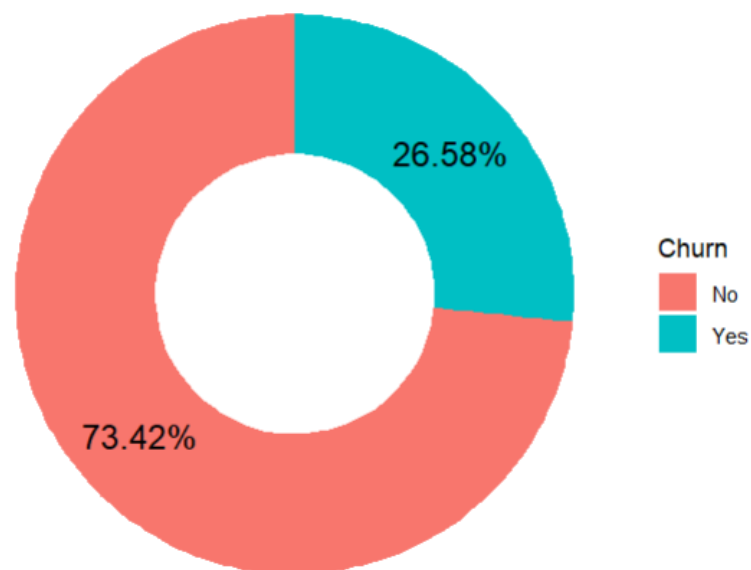


R code:

```
> ggplot(churn_df, aes(x = 2, y = Count, fill = Churn)) +  
+   geom_col(width = 1) +  
+   coord_polar(theta = "y") +  
+   xlim(0.5, 2.5) +  
+   geom_text(  
+     aes(label = paste0(Percentage, "%")),  
+     position = position_stack(vjust = 0.5),  
+     size = 5  
+   ) +  
+   theme_void() +  
+   labs(title = "Customer Churn Distribution")
```

Output:

Customer Churn Distribution



Result

The churn distribution analysis revealed that out of **7,043 customers**, **5,174 customers (73.46%)** continued using the telecom services, while **1,869 customers (26.54%)** discontinued the services and were classified as churned customers.

The results indicate that approximately **one out of every four customers** left the company, highlighting customer attrition as a significant business concern. This level of churn can negatively

impact revenue and increase customer acquisition costs, emphasizing the importance of identifying the factors influencing churn and implementing effective customer retention strategies.

- Total Customers: **7,043**
- Retained Customers: **5,174 (73.46%)**
- Churned Customers: **1,869 (26.54%)**
- Approximately **26.5% of customers churned**, indicating a substantial opportunity for customer retention initiatives.

4. Statistical Hypothesis Testing

R code:

```
> table(churn_data$Contract)
```

Output:

Month-to-month	One year	Two year
3875	1472	1685

R code:

```
> table(churn_data$Contract, churn_data$Churn)
```

Output:

	No	Yes
Month-to-month	2220	1655
One year	1306	166
Two year	1637	48

R code

```
> chisq.test(table(churn_data$Contract, churn_data$Churn))
```

Output:

Pearson's Chi-squared test

data: table(churn_data\$Contract, churn_data\$Churn)

X-squared = 1179.5, df = 2, p-value < 2.2e-16

Result

A Chi-Square Test of Independence was performed to examine the relationship between **Contract Type** and **Customer Churn**.

The results showed a highly significant association between the two variables:

- **Chi-Square Statistic (χ^2) = 1179.5**

- **Degrees of Freedom (df) = 2**
- **p-value < 0.001**

Since the p-value is less than the significance level of 0.05, the null hypothesis was rejected. This indicates that **customer churn is significantly associated with contract type**.

The contingency table revealed that customers with **Month-to-Month contracts** experienced the highest number of churn cases (1,655 customers), whereas customers with **One-Year** and **Two-Year contracts** showed substantially lower churn counts.

Conclusion

- Contract type has a significant impact on customer churn.
- Customers on **Month-to-Month contracts** are more likely to leave the company.
- Customers on **One-Year** and especially **Two-Year contracts** are more likely to remain with the company.
- Encouraging customers to switch from Month-to-Month plans to long-term contracts could help improve customer retention and reduce churn rates.

Business Insight

The analysis suggests that long-term contractual commitments play a crucial role in customer retention. Telecom companies can reduce churn by offering incentives, discounts, or loyalty benefits that encourage customers to adopt longer-duration contracts.

R code:

```
> aggregate(
+   MonthlyCharges ~ Churn,
+   data = churn_data,
+   mean
+ )
```

Output:

```
Churn MonthlyCharges
1 No      61.30741
2 Yes     74.44133
```

R code:

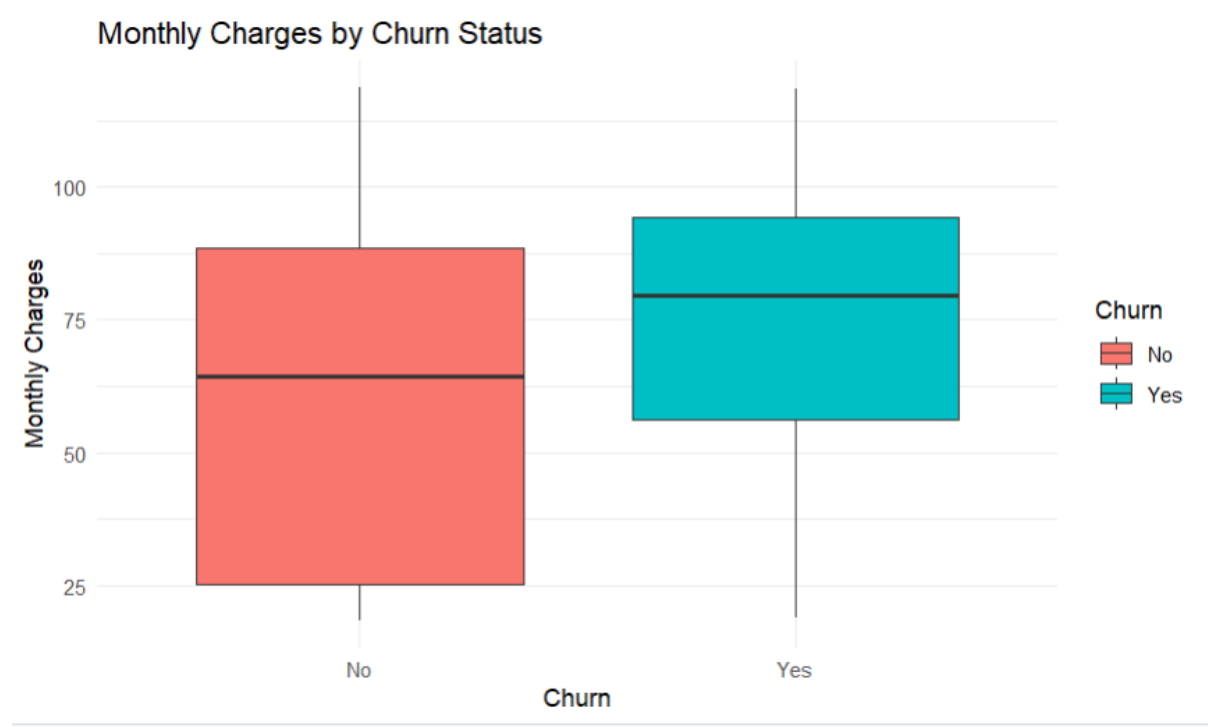
```
> library(ggplot2)
>
> ggplot(churn_data,
+   aes(x = Churn,
```

```

+   y = MonthlyCharges,
+   fill = Churn)) +
+   geom_boxplot() +
+   labs(
+     title = "Monthly Charges by Churn Status",
+     x = "Churn",
+     y = "Monthly Charges"
+   ) +
+   theme_minimal()

```

Output:



R code:

```

> t.test(
+   MonthlyCharges ~ Churn,
+   data = churn_data
+ )

```

Output:

Welch Two Sample t-test

data: MonthlyCharges by Churn

$t = -18.341$, $df = 4139.7$, $p\text{-value} < 2.2e-16$

alternative hypothesis: true difference in means between group No and group Yes is not equal to 0

95 percent confidence interval:

-14.53786 -11.72998

sample estimates:

mean in group No mean in group Yes

61.30741 74.44133

Result

A boxplot was created to compare the distribution of **Monthly Charges** between churned and non-churned customers. The visualization indicated that customers who churned generally had higher monthly charges than those who remained with the company.

To statistically verify this difference, a **Welch Two-Sample t-test** was performed.

The test results were:

- **t-statistic** = -18.341
- **p-value** < 0.001
- **95% Confidence Interval** = (-14.54, -11.73)

Average monthly charges were:

Churn Status Mean Monthly Charges

No 61.31

Yes 74.44

Since the p-value is significantly less than 0.05, the null hypothesis was rejected, indicating a statistically significant difference in monthly charges between churned and retained customers.

Conclusion

- Customers who churned paid significantly higher monthly charges than customers who remained with the company.
- The average monthly charge for churned customers (**74.44**) was considerably higher than that of retained customers (**61.31**).
- Monthly charges are an important factor influencing customer churn.

Business Insight

The findings suggest that customers facing higher monthly service costs are more likely to discontinue their subscriptions. Telecom companies should closely monitor customers with high

monthly charges and consider offering personalized discounts, loyalty rewards, or alternative pricing plans to improve customer retention and reduce churn.

R code:

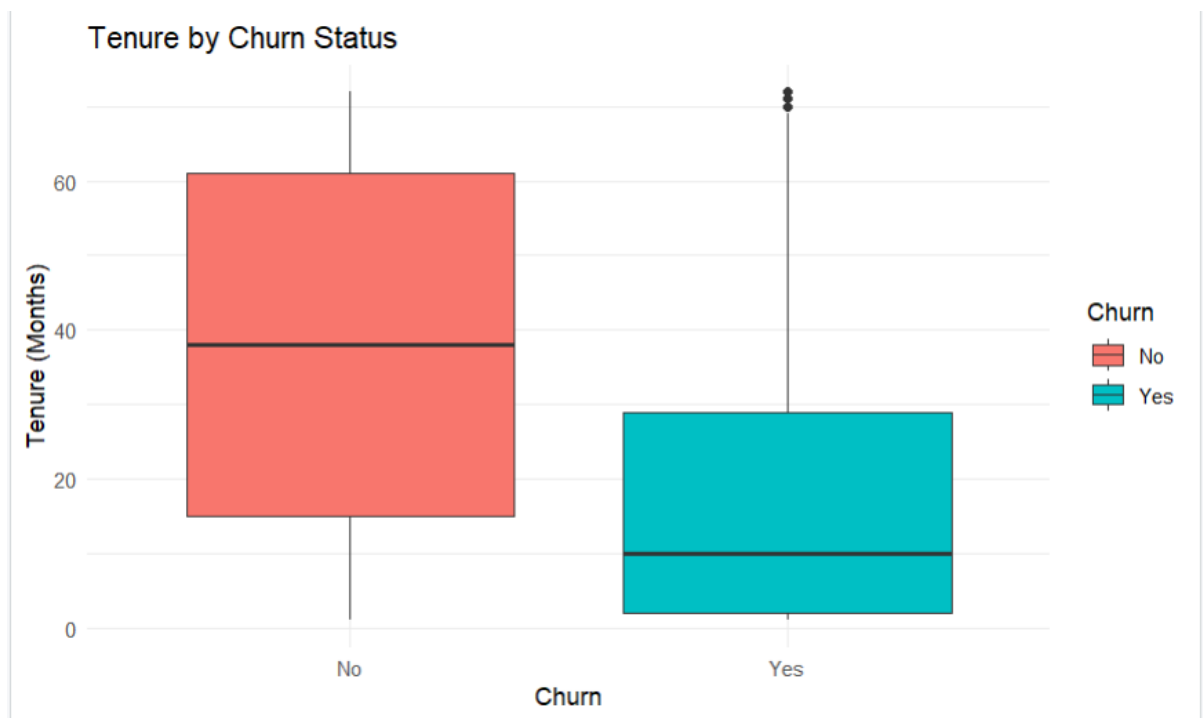
```
> aggregate(  
+   tenure ~ Churn,  
+   data = churn_data,  
+   mean  
+ )
```

Output:

```
Churn tenure  
1 No 37.65001  
2 Yes 17.97913
```

R Code:

```
> ggplot(churn_data,  
+   aes(x = Churn,  
+       y = tenure,  
+       fill = Churn)) +  
+   geom_boxplot() +  
+   labs(  
+     title = "Tenure by Churn Status",  
+     x = "Churn",  
+     y = "Tenure (Months)"  
+   ) +  
+   theme_minimal()
```

Output:**R Code:**

```
> t.test(  
+   tenure ~ Churn,  
+   data = churn_data  
+ )
```

Output:

Welch Two Sample t-test

data: tenure by Churn

t = 34.972, df = 4045.5, p-value < 2.2e-16

alternative hypothesis: true difference in means between group No and group Yes is not equal to 0

95 percent confidence interval:

18.56811 20.77364

sample estimates:

mean in group No mean in group Yes

37.65001 17.97913

Result

A boxplot was used to compare the distribution of **customer tenure** between churned and non-churned customers. The visualization showed that customers who remained with the company generally had a much longer tenure than customers who churned.

To determine whether this difference was statistically significant, a **Welch Two-Sample t-test** was conducted.

The test results were:

- **t-statistic** = 34.972
- **p-value** < 0.001
- **95% Confidence Interval** = (18.57, 20.77)

The average tenure for each group was:

Churn Status Mean Tenure (Months)

No	37.65
Yes	17.98

Since the p-value is significantly less than 0.05, the null hypothesis was rejected, indicating a statistically significant difference in tenure between churned and retained customers.

Conclusion

- Customers who remained with the company had substantially longer tenures than customers who churned.
- The average tenure of retained customers (**37.65 months**) was approximately **20 months higher** than that of churned customers (**17.98 months**).
- Customer tenure is a strong predictor of churn behavior.

Business Insight

The findings suggest that customer churn is more common among newer customers. Customers with shorter relationships with the company are significantly more likely to discontinue services. Therefore, telecom companies should focus on improving customer onboarding, engagement, and retention strategies during the early stages of the customer lifecycle to reduce churn and increase long-term customer loyalty.

Key Findings

The analysis identified several important factors associated with customer churn in the telecom industry:

1. High Customer Churn Rate

Out of 7,043 customers, 1,869 customers (26.54%) discontinued the service, indicating that approximately one in four customers churned.

2. Contract Type Strongly Influences Churn

The Chi-Square test revealed a statistically significant association between contract type and churn ($\chi^2 = 1179.5$, $p < 0.001$).

Customers with Month-to-Month contracts exhibited substantially higher churn rates compared to customers with One-Year and Two-Year contracts.

3. Higher Monthly Charges Increase Churn Risk

The average monthly charge for churned customers was \$74.44, compared to \$61.31 for retained customers.

The Welch Two-Sample t-test confirmed that this difference was statistically significant ($p < 0.001$).

4. Customer Tenure Significantly Impacts Churn

Retained customers had an average tenure of 37.65 months, while churned customers had an average tenure of only 17.98 months.

The difference was statistically significant ($p < 0.001$), indicating that newer customers are considerably more likely to leave the company.

5. Customer Retention Improves Over Time

Customers who remain with the company for longer periods tend to develop stronger loyalty and are less likely to churn.

Business Recommendations

Based on the findings, the following strategies are recommended:

Recommendation 1: Encourage Long-Term Contracts

Since Month-to-Month customers have the highest churn rates, the company should provide:

- Contract upgrade discounts
- Loyalty rewards
- Long-term subscription benefits

to encourage migration to One-Year and Two-Year plans.

Recommendation 2: Monitor High-Billing Customers

Customers paying higher monthly charges are more likely to churn.

The company should:

- Offer personalized pricing plans
- Provide targeted discounts
- Introduce value-added services

for customers with high monthly bills.

Recommendation 3: Strengthen Early Customer Engagement

Customers with shorter tenure represent the highest-risk segment.

The company should:

- Improve onboarding experiences
- Conduct follow-up customer satisfaction surveys
- Offer retention incentives during the first year

to improve long-term retention.

Recommendation 4: Develop a Churn Monitoring System

A predictive analytics framework should be implemented to continuously identify customers at risk of churn and trigger proactive retention campaigns.

Project Conclusion

This project analyzed customer churn behavior using statistical techniques and exploratory data analysis on the IBM Telco Customer Churn dataset.

The findings revealed that contract type, monthly charges, and customer tenure are significant factors influencing customer churn. Customers on Month-to-Month contracts, customers paying higher monthly charges, and customers with shorter tenure were found to have substantially higher probabilities of leaving the company.

The statistical tests confirmed that these relationships were highly significant, providing strong evidence that customer churn is not random but is influenced by measurable customer characteristics.

Overall, the study demonstrates how statistical analysis can be used to identify churn drivers, support business decision-making, and develop effective customer retention strategies. By focusing on high-risk customer segments and implementing targeted retention initiatives, telecom companies can reduce churn, improve customer satisfaction, and enhance long-term profitability.